

homework 1

chl

Exercise of Chapter 1 in the book. [Shakarchi,Stein]Complex Analysis(2003)
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7.

The family of mappings introduced here plays an important role in complex analysis. These mappings, sometimes called Blaschke factors, will reappear in various applications in later chapters.

(a) Let z, w be two complex numbers such that $\bar{z}w \neq 1$. Prove that

$$\left| \frac{w - z}{1 - \bar{w}z} \right| < 1 \quad \text{if } |z| < 1 \text{ and } |w| < 1,$$

and also that

$$\left| \frac{w - z}{1 - \bar{w}z} \right| = 1 \quad \text{if } |z| = 1 \text{ or } |w| = 1.$$

[Hint: Why can one assume that z is real? It then suffices to prove that

$$(r - w)(r - \bar{w}) \leq (1 - rw)(1 - r\bar{w})$$

with equality for appropriate r and $|w|$.]

Proof. Since $\left| \frac{w-z}{1-\bar{w}z} \right| < 1 \iff |w-z|^2 - |1-\bar{w}z|^2 < 0$,

$$\begin{aligned} |w-z|^2 - |1-\bar{w}z|^2 &= |w|^2 + |z|^2 - w\bar{z} - z\bar{w} - (1 + |w|^2|z|^2 - \bar{w}z - z\bar{w}) \\ &= -(1-|w|)(1-|z|) < 0 \iff |z| < 1 \text{ and } |w| < 1 \end{aligned}$$

There is equality if and only if $|z| = 1$ or $|w| = 1$. □

(b) Prove that for a fixed w in the unit disc D , the mapping

$$F : z \mapsto \frac{w-z}{1-\bar{w}z}$$

satisfies the following conditions:

- (i) F maps the unit disc to itself (that is, $F : D \rightarrow D$), and is holomorphic.
- (ii) F interchanges 0 and w , namely $F(0) = w$ and $F(w) = 0$.
- (iii) $|F(z)| = 1$ if $|z| = 1$.
- (iv) $F : D \rightarrow D$ is bijective. [Hint: Calculate $F \circ F$.]

Proof. 1. $D = \{z : |z| < 1\}$. $z \in D, w \in D \implies |z| < 1, |w| < 1 \implies |F(z)| < 1 \implies F(z) \in D$.
 $\forall z \in D, |1-\bar{w}z| \leq 1-|w||z| \in (0,1) \implies 1-\bar{w}z \neq 0$,

$$F'(z) = \frac{-(1-\bar{w}z) - (w-z)(-\bar{w})}{(1-\bar{w}z)^2} = \frac{|w|^2-1}{(1-\bar{w}z)^2}$$

$\implies f$ is holomorphic on D .

2.

$$F(0) = \frac{w-0}{1-0} = w \quad F(w) = \frac{w-w}{1-\bar{w}w} = 0$$

3.

$$|F(z)| = \left| \frac{w-z}{1-\bar{w}z} \right|$$

In (a) we know that if $|z| = 1$, then $|F(z)| = 1$.

4.

$$\begin{aligned} F \circ F(z) &= F\left(\frac{w-z}{1-\bar{w}z}\right) = \frac{w - \frac{w-z}{1-\bar{w}z}}{1 - \bar{w}\frac{w-z}{1-\bar{w}z}} = \frac{w - |w|^2z - w + z}{1 - \bar{w}z - |w|^2 + \bar{w}z} \\ &= \frac{z(1-|w|^2)}{1-|w|^2} = z \implies F \circ F = 1_D \end{aligned}$$

This means f is injective and surjective, thus, f is bijective. □

8.

Suppose U and V are open sets in the complex plane. Prove that if $f : U \rightarrow V$ and $g : V \rightarrow \mathbb{C}$ are two functions that are differentiable (in the real sense, that is, as functions of the two real variables x and y), and $h = g \circ f$, then

$$\frac{\partial h}{\partial z} = \frac{\partial g}{\partial z} \frac{\partial f}{\partial z} + \frac{\partial g}{\partial \bar{z}} \frac{\partial \bar{f}}{\partial z}$$

and

$$\frac{\partial h}{\partial \bar{z}} = \frac{\partial g}{\partial z} \frac{\partial f}{\partial \bar{z}} + \frac{\partial g}{\partial \bar{z}} \frac{\partial \bar{f}}{\partial \bar{z}}.$$

This is the complex version of the chain rule.

Proof. Let $z = x + iy$, and write

$$f(z) = u(x, y) + iv(x, y).$$

Set $w = f(z)$ and $h(z) = g(f(z))$. Since f and g are differentiable in the real sense, the usual chain rule gives

$$h_x = g_u u_x + g_v v_x, \quad h_y = g_u u_y + g_v v_y.$$

First compute

$$\frac{\partial h}{\partial z} = \frac{1}{2}(h_x - ih_y) = \frac{1}{2}[(g_u u_x + g_v v_x) - i(g_u u_y + g_v v_y)] = \frac{1}{2}[g_u(u_x - iu_y) + g_v(v_x - iv_y)].$$

Since

$$g_u = g_w + g_{\bar{w}}, \quad g_v = i(g_w - g_{\bar{w}}),$$

we obtain

$$\frac{\partial h}{\partial z} = \frac{1}{2}g_w[(u_x - iu_y) + i(v_x - iv_y)] + \frac{1}{2}g_{\bar{w}}[(u_x - iu_y) - i(v_x - iv_y)].$$

Since

$$(u_x - iu_y) + i(v_x - iv_y) = 2f_z, \quad (u_x - iu_y) - i(v_x - iv_y) = 2\bar{f}_z,$$

we have

$$\frac{\partial h}{\partial z} = g_w f_z + g_{\bar{w}} \bar{f}_z = \frac{\partial g}{\partial z} \frac{\partial f}{\partial z} + \frac{\partial g}{\partial \bar{z}} \frac{\partial \bar{f}}{\partial z}.$$

Similarly,

$$\frac{\partial h}{\partial \bar{z}} = \frac{1}{2}(h_x + ih_y) = \frac{1}{2}[g_u(u_x + iu_y) + g_v(v_x + iv_y)].$$

Substituting the same expressions for g_u and g_v gives

$$\frac{\partial h}{\partial \bar{z}} = \frac{1}{2}g_w \left[(u_x + iu_y) + i(v_x + iv_y) \right] + \frac{1}{2}g_{\bar{w}} \left[(u_x + iu_y) - i(v_x + iv_y) \right].$$

Noting that

$$(u_x + iu_y) + i(v_x + iv_y) = 2f_{\bar{z}}, \quad (u_x + iu_y) - i(v_x + iv_y) = 2\bar{f}_{\bar{z}},$$

we obtain

$$\frac{\partial h}{\partial \bar{z}} = g_w f_{\bar{z}} + g_{\bar{w}} \bar{f}_{\bar{z}} = \frac{\partial g}{\partial z} \frac{\partial f}{\partial \bar{z}} + \frac{\partial g}{\partial \bar{z}} \frac{\partial \bar{f}}{\partial \bar{z}}.$$

□

9.

Show that in polar coordinates, the Cauchy-Riemann equations take the form

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \quad \text{and} \quad \frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}.$$

Use these equations to show that the logarithm function defined by

$$\log z = \log r + i\theta \quad \text{where } z = re^{i\theta} \text{ with } -\pi < \theta < \pi$$

is holomorphic in the region $r > 0$ and $-\pi < \theta < \pi$.

Proof. Let

$$f(z) = u(x, y) + iv(x, y), \quad x = r \cos \theta, \quad y = r \sin \theta.$$

By the chain rule,

$$\begin{aligned} u_r &= u_x \frac{\partial x}{\partial r} + u_y \frac{\partial y}{\partial r} = u_x \cos \theta + u_y \sin \theta, \\ u_\theta &= u_x \frac{\partial x}{\partial \theta} + u_y \frac{\partial y}{\partial \theta} = u_x(-r \sin \theta) + u_y(r \cos \theta). \end{aligned}$$

Similarly,

$$v_r = v_x \cos \theta + v_y \sin \theta,$$

$$v_\theta = v_x(-r \sin \theta) + v_y(r \cos \theta).$$

Substituting

$$u_x = v_y, \quad u_y = -v_x,$$

gives

$$\begin{aligned} u_r &= u_x \cos \theta + u_y \sin \theta = v_y \cos \theta - v_x \sin \theta = \frac{1}{r}(-rv_x \sin \theta + rv_y \cos \theta) = \frac{1}{r}v_\theta, \\ \frac{1}{r}u_\theta &= -u_x \sin \theta + u_y \cos \theta = -v_y \sin \theta - v_x \cos \theta = -(v_x \cos \theta + v_y \sin \theta) = -v_r. \end{aligned}$$

Thus in polar coordinates, the Cauchy-Riemann equations take the form

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}.$$

Now consider

$$\log z = \log r + i\theta \quad \text{where } z = re^{i\theta}, \quad -\pi < \theta < \pi.$$

Here

$$u(r, \theta) = \log r, \quad v(r, \theta) = \theta.$$

We have

$$u_r = \frac{1}{r}, \quad u_\theta = 0, \quad v_r = 0, \quad v_\theta = 1.$$

Then

$$\begin{aligned} u_r &= \frac{1}{r} = \frac{1}{r}v_\theta \\ \frac{1}{r}u_\theta &= 0 = -v_r \end{aligned}$$

So the polar form of the Cauchy-Riemann equations is satisfied for $r > 0$ and $-\pi < \theta < \pi$. Since u and v have continuous derivatives there, then $\log z$ is holomorphic in the region

$$r > 0, \quad -\pi < \theta < \pi.$$

□

10.

Show that

$$4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} = 4 \frac{\partial}{\partial \bar{z}} \frac{\partial}{\partial z} = \Delta,$$

where Δ is the Laplacian

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}.$$

Proof. We know that

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial x} \frac{\partial x}{\partial z} + \frac{\partial}{\partial y} \frac{\partial y}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right)$$

$$\frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial x} \frac{\partial x}{\partial \bar{z}} + \frac{\partial}{\partial y} \frac{\partial y}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right)$$

Then

$$4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} = 4 \cdot \frac{1}{2} \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right) \cdot \frac{1}{2} \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right) = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \Delta$$

$$4 \frac{\partial}{\partial \bar{z}} \frac{\partial}{\partial z} = 4 \cdot \frac{1}{2} \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right) \cdot \frac{1}{2} \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right) = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \Delta$$

□

16.

Determine the radius of convergence of the series $\sum_{n=1}^{\infty} a_n z^n$ when:

(a) $a_n = (\log n)^2$

Proof.

$$R = \frac{1}{\limsup_{n \rightarrow \infty} (\log n)^{\frac{2}{n}}} = \frac{1}{\exp(\limsup_{n \rightarrow \infty} \frac{2 \log \log n}{n})} = \frac{1}{e^0} = 1$$

□

(b) $a_n = n!$

Proof.

$$R = \frac{1}{\limsup_{n \rightarrow \infty} (n!)^{\frac{1}{n}}} = \frac{1}{\exp(\limsup_{n \rightarrow \infty} \frac{\sum_{k=1}^n \log k}{n})} = \frac{1}{\exp(\limsup_{n \rightarrow \infty} \frac{\log n}{1})} = 0$$

□

$$(c) \ a_n = \frac{n^2}{4^n + 3n}$$

Proof.

$$R = \frac{1}{\limsup_{n \rightarrow \infty} \left(\frac{n^2}{4^n + 3n}\right)^{\frac{1}{n}}} = \frac{1}{\limsup_{n \rightarrow \infty} \left(\frac{n^2}{4^n}\right)^{\frac{1}{n}}} = \frac{4}{\limsup_{n \rightarrow \infty} (n)^{\frac{2}{n}}} = 4$$

□

$$(d) \ a_n = \frac{(n!)^3}{(3n)!} \quad [\text{Hint: Use Stirling's formula, which says that } n! \sim cn^{n+\frac{1}{2}}e^{-n} \text{ for some } c > 0.]$$

Proof.

$$R = \frac{1}{\limsup_{n \rightarrow \infty} \left(\frac{(n!)^3}{(3n)!}\right)^{\frac{1}{n}}} = \frac{1}{\limsup_{n \rightarrow \infty} \left(\frac{(cn^{n+\frac{1}{2}}e^{-n})^3}{c(3n)^{3n+\frac{1}{2}}e^{-3n}}\right)^{\frac{1}{n}}} = 27$$

□

(e) Find the radius of convergence of the hypergeometric series

$$F(\alpha, \beta, \gamma; z) = 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha+1) \cdots (\alpha+n-1)\beta(\beta+1) \cdots (\beta+n-1)}{n!\gamma(\gamma+1) \cdots (\gamma+n-1)} z^n.$$

Here $\alpha, \beta \in \mathbb{C}$ and $\gamma \neq 0, -1, -2, \dots$

Proof.

$$a_n = \frac{\alpha(\alpha+1) \cdots (\alpha+n-1)\beta(\beta+1) \cdots (\beta+n-1)}{n!\gamma(\gamma+1) \cdots (\gamma+n-1)}.$$

Then

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(\alpha+n)(\beta+n)}{(n+1)(\gamma+n)} \right| = 1.$$

Therefore,

$$R = \frac{1}{\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|} = 1.$$

□

(f) Find the radius of convergence of the Bessel function of order r :

$$J_r(z) = \left(\frac{z}{2}\right)^r \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+r)!} \left(\frac{z}{2}\right)^{2n},$$

where r is a positive integer.

Proof.

$$J_r(z) = \left(\frac{z}{2}\right)^r \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+r)!} \left(\frac{z}{2}\right)^{2n} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{2n+r} n!(n+r)!} z^{2n+r}.$$

Let

$$a_n = \frac{(-1)^n}{2^{2n+r} n!(n+r)!}.$$

Then

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{1}{4(n+1)(n+r+1)} = 0$$

Therefore,

$$\limsup_{n \rightarrow \infty} a_n^{\frac{1}{n}} = 0 \quad R = \infty$$

□

23.

Consider the function f defined on $\mathbb{R}$ by

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0, \\ e^{-1/x^2} & \text{if } x > 0. \end{cases}$$

Prove that f is indefinitely differentiable on $\mathbb{R}$, and that $f^{(n)}(0) = 0$ for all $n \geq 1$. Conclude that f does not have a converging power series expansion $\sum_{n=0}^{\infty} a_n x^n$ for x near the origin.

Proof. For $x < 0$, $f(x) = 0$, so f is C^∞ there. For $x > 0$, $f(x) = e^{-1/x^2}$, so it is trivial that f is also C^∞ there. Thus we only need to study the behavior at $x = 0$.

Suppose that for every $n \geq 0$, there exists a polynomial P_n such that for $x > 0$,

$$f^{(n)}(x) = P_n(1/x)e^{-1/x^2}.$$

We prove this by induction. For $n = 0$, this is trivial with $P_0(t) = 1$. Suppose

$$f^{(n)}(x) = P_n(1/x)e^{-1/x^2}.$$

Then for $x > 0$,

$$f^{(n+1)}(x) = \frac{d}{dx}(P_n(1/x)e^{-1/x^2}) = -\frac{1}{x^2}P'_n(1/x)e^{-1/x^2} + P_n(1/x)\frac{2}{x^3}e^{-1/x^2}.$$

which is of the form $P_{n+1}(1/x)e^{-1/x^2}$ for some polynomial P_{n+1} .

Therefore,

$$\lim_{x \rightarrow 0^+} \frac{f^{(n)}(x) - 0}{x - 0} = \lim_{x \rightarrow 0^+} \frac{1/x P_n(1/x)}{e^{1/x^2}} = \lim_{t \rightarrow \infty} \frac{t P_n(t)}{e^{t^2}} = 0$$

Hence, $\forall n, f^{(n)}(0) = 0, f \in C^\infty(\mathbb{R})$.

Now suppose f had a convergent power series expansion near 0:

$$f(x) = \sum_{n=0}^{\infty} a_n x^n.$$

Then, since a convergent power series is equal to its Taylor series,

$$a_n = \frac{f^{(n)}(0)}{n!} = 0 \quad \text{for all } n \geq 0.$$

Hence the series would be identically zero near 0, so $f(x) \equiv 0$ for all x sufficiently close to 0. But for $x > 0$,

$$f(x) = e^{-1/x^2} > 0,$$

$\implies$ *contradiction*! Therefore f does not have a convergent power series expansion for x near the origin. $\square$

25.

The next three calculations provide some insight into Cauchy's theorem, which we treat in the next chapter.

(a) Evaluate the integrals

$$\int_{\gamma} z^n dz$$

for all integers n . Here γ is any circle centered at the origin with the positive (counterclockwise) orientation.

Proof. For any circle with radius R and center the origin,

$$\begin{aligned} \int_{\gamma} z^n dz &= \int_0^{2\pi} R^n e^{in\theta} R i e^{i\theta} d\theta = R^{n+1} i \int_0^{2\pi} e^{i(n+1)\theta} d\theta \\ &= R^{n+1} i \left[\frac{e^{i(n+1)\theta}}{i(n+1)} \right]_0^{2\pi} = 0 \end{aligned}$$

□

(b) Same question as before, but with γ any circle not containing the origin.

Proof. For any circle with radius R and center z_0 ($R > |z_0|$),

$$\begin{aligned} \int_{\gamma} z^n dz &= \int_0^{2\pi} (z_0 + R e^{i\theta})^n R i e^{i\theta} d\theta \\ &= \int_0^{2\pi} \left(\sum_{k=0}^n \binom{n}{k} z_0^{n-k} R^k e^{ik\theta} \right) R i e^{i\theta} d\theta \\ &= R i \sum_{k=0}^n \binom{n}{k} z_0^{n-k} R^k \int_0^{2\pi} e^{i(k+1)\theta} d\theta \\ &= R i \sum_{k=0}^n \binom{n}{k} z_0^{n-k} R^k \left[\frac{e^{i(k+1)\theta}}{i(k+1)} \right]_0^{2\pi} \\ &= 0 \end{aligned}$$

□

(c) Show that if $|a| < r < |b|$, then

$$\int_{\gamma} \frac{1}{(z-a)(z-b)} dz = \frac{2\pi i}{a-b},$$

where γ denotes the circle centered at the origin, of radius r , with the positive orientation.

Proof.

$$\int_{\gamma} \frac{1}{(z-a)(z-b)} dz = \frac{1}{a-b} \left(\int_{\gamma} \frac{1}{z-a} dz - \int_{\gamma} \frac{1}{z-b} dz \right)$$

Since $|a| < r$, the point a lies inside γ . Then

$$\begin{aligned}
\int_{\gamma} \frac{1}{z-a} dz &= \int_0^{2\pi} \frac{1}{re^{i\theta} - a} ire^{i\theta} d\theta \\
&= \int_0^{2\pi} \frac{i}{1 - \frac{a}{re^{i\theta}}} d\theta \\
&= \int_0^{2\pi} i \sum_{n=0}^{\infty} \left(\frac{a}{re^{i\theta}} \right)^n d\theta (\text{converges}) \\
&= i \sum_{n=0}^{\infty} \frac{a^n}{r^n} \int_0^{2\pi} e^{-in\theta} d\theta (= 0 \text{ if } n \geq 1) \\
&= i \int_0^{2\pi} 1 d\theta \\
&= 2\pi i.
\end{aligned}$$

Since $|b| > r$, the point b lies outside γ . Similarly,

$$\begin{aligned}
\int_{\gamma} \frac{1}{z-b} dz &= \int_0^{2\pi} \frac{1}{re^{i\theta} - b} ire^{i\theta} d\theta \\
&= - \int_0^{2\pi} \frac{ire^{i\theta}}{b} \frac{1}{1 - \frac{re^{i\theta}}{b}} d\theta \\
&= - \int_0^{2\pi} \frac{ire^{i\theta}}{b} \sum_{n=0}^{\infty} \left(\frac{re^{i\theta}}{b} \right)^n d\theta \\
&= -i \sum_{n=0}^{\infty} \frac{r^{n+1}}{b^{n+1}} \int_0^{2\pi} e^{i(n+1)\theta} d\theta \\
&= 0.
\end{aligned}$$

Therefore

$$\int_{\gamma} \frac{1}{(z-a)(z-b)} dz = \frac{1}{a-b} (2\pi i - 0) = \frac{2\pi i}{a-b}.$$

□